A Generic Dynamic Logic for Program Reasoning based on Operational Semantics

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Abstract. Dynamic logic is a valuable formalism that has many applications in ensuring safety-critical systems. We present a novel theory of parameterized dynamic logic, namely DL_p, for specifying and reasoning about general program models based on their operational semantics. Different from most dynamic logics that deal with regular expressions or a particular type of models, DL_p allows arbitrary forms of programs and formulas according to interested domains. It provides a language-independent efficient proof calculus under dynamic-logic settings, supporting a symbolic-execution-based reasoning with a general notion of labels for capturing program configurations. To admits certain infinite proof deductions caused by loop programs, we adapt the cyclic proof approach to the theory of DL_p by building a cyclic proof structure special for DL_p. The soundness of DL_p is analyzed and formally proved. A case study displays an instantiation of DL_p in particular domains, demonstrating the potential usage of DL_p in program verifications.

Keywords: Dynamic Logic · Program Deduction · Verification Framework · Cyclic Proof · Operational Semantics · Symbolic Execution

1 Introduction

Background and Motivations. Dynamic logic [21] has proven to be a valuable formalism for specifying and reasoning about different types of *programs* — a term used here in a broad sense beyond traditional computer programs. It has been successfully applied to domains such as process algebras [6], programming languages [5], synchronous systems [49,48], hybrid systems [34,35] and probabilistic systems [33,27]. These theories have inspired the developments of related verification tools like [38,42,36,49,14] for safety-critical systems. Recent developments of dynamic logic include a range of extensions aimed at addressing contemporary challenges, such as ensuring the safety of blockchain systems [24], describing hyperproperties [20], and verifying quantum computations [45,14]. It also has attracted attention as a promising framework for "incorrectness reasoning", as explored recently in [32,50].

As an extension of modal logic, dynamic logic embeds a program α into the modality $\Box$ of a modal formula $\Box\phi$ as a form: $[\alpha]\phi$, meaning that after all executions of α, formula ϕ holds. This so-called "dynamic formula" allows

multiple and nested modalities in forms like $[\alpha]\phi \to \langle\beta\rangle\psi$ and $[\alpha]\langle\beta\rangle\phi$ (where $\langle\cdot\rangle$ is the dual modality of $[\cdot]$), being able to directly express and reason about more complex program properties than Hoare triple $\{\phi\}\alpha\{\psi\}$ in Hoare logics (e.g. [23,39]). Particularly, when restricting a program α to be deterministic, formulas $\phi \to [\alpha]\psi$ and $\phi \to \langle\alpha\rangle\psi$ exactly capture the meanings of partial and total correctness of program α respectively in Hoare logics.

Standard dynamic logics [21] (as well as Hoare logics such as [23,39]) are in explicit forms. When applying them to a certain system model or computer program, new theories are required to adapt the target languages. For instance, to apply first-order dynamic logic (FODL) to verification of Java programs, many primitives in Java and inference rules special for structures of Java need to be added into the FODL theory (cf. [5]). Otherwise, additional language transformations are inevitable, causing loses of critical program structural information during deductions. However, many languages in reality, such as AADL, UML/MARTE, Esterel [7], Java, C, etc., often have complex structures. Building a specific logic theory for them requires a lot of work. Moreover, the proof system of a logic theory is often error prone, thus requiring validation of its soundness (and even completeness), which also can be costly. One famous example is that Verifiable C [3] spends nearly 40,000 lines of Coq code to define and validate its logic theory for C programming language based on separation logic [39].

Another main issue is that the proof systems of dynamic logics are often based on programs' denotational semantics (cf. [21]). For most interested computer systems and programs, on the other hand, operational semantics — that describes a program is symbolically executed from one configuration to another — is in their nature. In these cases, the consistency from denotational semantics to operational semantics has to be validated (cf. [40,30]). This thus increases the burden of using dynamic logics. Moreover, the denotational semantics of some languages are not compositional. Typical examples are neural networks [19], automata-based system models like [47] and some concurrent programming languages, such as synchronous languages Esterel [8] and Quartz [18]. For these models, reasoning based on symbolic executions is more direct, by avoiding unnecessary language transformations. [18] shows an example of non-compositional derivations of Quartz programs that rely on heavy transformations.

This paper focuses on a theoretical solution of the above two problems in existing dynamic-logic theories. We present a novel dynamic logic called "parameterized dynamic logic" (abbreviated as DL_p) for reasoning about programs based on their operational semantics. Generally speaking, DL_p provides a language-independent verification framework in which program deductions rely solely on program transitions of a universal form: $(\alpha, \sigma) \longrightarrow (\alpha', \sigma')$ (for arbitrary $\alpha, \alpha', \sigma, \sigma'$). Instead of carefully designing rules according to programs' denotational semantics, as we will see, inference rules for program transitions are natural from programs' operational semantics and are much less error prone. So in most cases we can trust them without any validations.

The methodology of reasoning by symbolically executing programs has been proposed and studied for years within different mathematical theories, among

them [40,41,43,12,30,28,22] are closest to our work. Except a few [31,22] (see Section 6 for a detailed comparison), most of the previous work has not yet addressed a similar approach under dynamic-logic settings, i.e., to provide an efficient logical calculus for deriving dynamic formulas $[\alpha]\phi$. In our opinion, it is valuable to fill in this gap.

Illustration of Main Idea. Informally, DL_p treats program α and formula ϕ of $[\alpha]\phi$ as 'parameters', while introducing 'labels' σ as program configurations to capture current program states for symbolic executions. The structures of α, ϕ and σ are irrelevant to the main skeleton of the verification framework. In DL_p, we derive a labelled DL_p dynamic formula $\sigma : [\alpha]\phi$ instead of $[\alpha]\phi$. When σ is general enough, $\sigma : [\alpha]\phi$ can capture the same meaning as $[\alpha]\phi$.

To see how we can benefit from deriving a labelled DL_p formula, consider a simple example. We want to prove a formula $\phi_1 =_{df} (x \geq 0 \rightarrow [x := x+1]x > 0)$ in FODL [37], where x is a variable ranging over integers $\mathbb{Z}$. Intuitively, formula ϕ_1 means that if $x \geq 0$ holds, then $x > 0$ holds after the assigning expression $x + 1$ to x. In FODL, to derive ϕ_1, we apply rule:

$$\frac{\phi[x/e]}{[x := e]\phi}\ {\scriptstyle (x:=e)}$$

for assignment on formula $[x := x+1]x > 0$ by substituting x of $x > 0$ with $x+1$, and obtain $x + 1 > 0$. Formula ϕ_1 thus becomes $\phi_1' =_{df} (x \geq 0 \rightarrow x + 1 > 0)$, which is true for any $x \in \mathbb{Z}$.

While in DL_p, formula ϕ_1 can be expressed as a form: $\psi_1 =_{df} (t \geq 0 \rightarrow \{x \mapsto t\} : [x := x + 1]x > 0)$, where formula $[x := x + 1]x > 0$ is labelled by configuration $\{x \mapsto t\}$, meaning "variable x stores value t" (with t a free variable). With a label explicitly showing up, to derive formula ψ_1, we instead directly perform a program transition of $x := x + 1$ as:

$$(x := x + 1, \{x \mapsto t\}) \longrightarrow (\downarrow, \{x \mapsto t + 1\}), \qquad (ex\ x := e)$$

which assign x with its current value t added by 1. Here $\downarrow$ indicates a program termination. Formula ψ_1 thus becomes $\psi_1' =_{df} (t \geq 0 \rightarrow \{x \mapsto t+1\} : x > 0)$, by replacing the part '$\{x \mapsto t\} : [x := x+1]$' with its execution result '$\{x \mapsto t+1\}$'. Formula $\{x \mapsto t + 1\} : x > 0$ exactly means $t + 1 > 0$, by replacing x with its current value $t + 1$. So from ψ_1' we obtain formula $t \geq 0 \rightarrow t + 1 > 0$, which is exactly formula ϕ_1' (modulo free variables).

In the above example, the form of the transition $(ex\ x := e)$ is closer to the operational semantics of the assignment than rule $(x := e)$. More importantly, this form is universal for all types of program transitions. By choosing different labels one can directly specify the rules for different languages. On the other hand, rule $(x := e)$ here is actually a special assignment rule in FODL. It may not be directly applied to other languages, for example, a Java statement $x :=$ *new* $C(...)$ that creates a new object of class C (cf. [5]).

Main Contributions & Challenges. In this paper, we firstly define the syntax and semantics of DL_p based on the standard theory of propositional dynamic logic (PDL) [17]. The difficulty of this part is to find a suitable way

to define the semantics of DL_p (Definition 4), since its ingredients are all unknown parameters. Following the conventions of defining a dynamic logic [21], we propose a so-called "program-labelled" (PL) Kripke structure (Definition 2) to support describing the operational semantics of programs in arbitrary forms. And we follow an approach similar in [15] to propose a labelled sequent calculus (Section 3.1) for symbolic-execution-based reasoning. But here the forms of labels in DL_p could be much more complex than in [15] (see Section 6).

After defining the logic theory, we build a cyclic verification framework for DL_p. Cyclic proof approach (cf. [11]) is a powerful technique to admit a certain type of infinite deductions (i.e. "cyclic proofs") caused by symbolic executions of programs with loop structures (Section 3.3). It has been attracting more and more attentions and recently has been applied for many logic theories such as [46,25,2]. We investigate this approach and adapt it to the theory of DL_p. The main challenges of this part are (1) the invention of a sound cyclic proof system, where the most critical work is to design rule (Sub) and the "substitutions of labels" (Definition 7); and (2) the construction of a cyclic proof structure, where the most key point is to define "progressive derivation traces" (Definition 9). At last, as one important work, we prove the soundness of our cyclic proof system (Section 5).

To sum up, the main contributions of this paper are three folds:

- We define the syntax and semantics of DL_p formulas.
- We build a labelled proof system for DL_p.
- We construct a sound cyclic proof system for DL_p and prove its soundness.

The rest of the paper is organized as follows. Section 2 defines the syntax and semantics of DL_p formulas. In Section 3, we propose a cyclic proof system for DL_p. In Section 4, we give an example of cyclic deductions of DL_p formulas. Section 5 analyzes and proves the soundness of DL_p. Section 6 introduces related work, while Section 7 makes a conclusion and discusses about future work.

2 Dynamic Logic DL_p

2.1 Propositional Dynamic Logic

We begin by introducing propositional dynamic logic (PDL) [17], the logic our construction of DL_p relies on.

The syntax of a PDL formula ϕ is given by simultaneous inductions on both programs and formulas as follows in BNF form:

$$\alpha =_{df} a \mid \phi? \mid \alpha\,;\alpha \mid \alpha \cup \alpha \mid \alpha^*$$
$$\phi =_{df} p \mid \neg\phi \mid \phi \wedge \phi \mid [\alpha]\phi,$$

where α is a regular expression with tests, often called a *regular program*. $a \in A$ is an atomic action of a symbolic set A; $\phi?$ is a test. If ϕ is true, then the program proceeds, otherwise, the program halts; $\alpha\,;\beta$ is a sequential program, meaning

that after program α proceeds and terminates, β continues proceeding; $\alpha \cup \beta$ is a choice program, it means that either α or β proceeds non-deterministically; α^* is a star program, which means that α is run for an arbitrary number $n \geq 0$ of times. p is an atomic formula, including the tautology *true*. We call a formula having dynamic forms a *dynamic formula*. Intuitively, formula $[\alpha]\phi$ means that after all executions of program α, formula ϕ holds.

The semantics of PDL is given based on a Kripke structure (cf. [21]) $M = (\mathcal{S}, \longrightarrow, \mathcal{I})$, where $\mathcal{S}$ is a set of *worlds*, representing program states; $\longrightarrow \subseteq \mathcal{S} \times A \times \mathcal{S}$ is a set of relations labeled by program actions; $\mathcal{I} : P \to \mathcal{P}(\mathcal{S})$ interprets each atomic PDL formula of set P to a set of worlds.

Given a Kripke structure M, based on the denotational semantics $[\![\cdot]\!]$ of regular programs, the semantics of PDL is given as a satisfaction relation $M, w \models \phi$ between a world w and a PDL formula ϕ. It is defined as follows by simultaneous inductions on both programs and formulas.

a. $[\![a]\!] =_{df} \{(w, w') \mid w \xrightarrow{a} w' \text{ on } M\}$;
b. $[\![\phi?]\!] =_{df} \{(w, w) \mid M, w \models \phi\}$;
c. $[\![\alpha\,;\beta]\!] =_{df} \{(w, w') \mid \exists w''.(w, w'') \in [\![\alpha]\!] \wedge (w'', w') \in [\![\beta]\!]\}$;
d. $[\![\alpha \cup \beta]\!] =_{df} \{(w, w') \mid (w, w') \in [\![\alpha]\!]\} \cup \{(w, w') \mid (w, w') \in [\![\beta]\!]\}$;
e. $[\![\alpha^*]\!] =_{df} \bigcup_{n=0}^{\infty} [\![\alpha^n]\!]$, where $\alpha^0 =_{df} \mathit{true}?$, $\alpha^n =_{df} \alpha^{n-1}\,;\alpha$ for any $n \geq 1$.

1. $M, w \models p$, if $w \in \mathcal{I}(p)$;
2. $M, w \models \neg\phi$, if $M, w \not\models \phi$;
3. $M, w \models \phi \wedge \psi$, if $M, w \models \phi$ and $M, w \models \psi$;
4. $M, w \models [\alpha]\phi$, if for all $(w, w') \in [\![\alpha]\!]$, $M, w' \models \phi$.

PDL studies formulas that are valid for all Kripke structures. Its proof system is complete (cf. [21]). First-order dynamic logic (FODL) [37] is obtained from PDL by specializing atomic actions a as assignments of the form $x := e$, where $x \in \mathit{Var}_{fodl}$ is a variable and e is an arithmetical expression of usually integer domain $\mathbb{Z}$. Many existing dynamic-logic theories [6,5,49,48,34,35,33,27,16] as mentioned above were developed based on FODL by extending regular programs with special language primitives and adding inference rules for these primitives.

2.2 Syntax and Semantics of DL_p

The theory of DL_p is built based on PDL, by allowing program α and formula ϕ of $[\alpha]\phi$ to be of arbitrary forms, only provided a restriction condition (Definition 10).

We assume two pre-defined sets $\mathbf{P}$ and $\mathbf{F}$, namely *parameters* of DL_p. $\mathbf{P}$ is a set of programs, in which we distinguish a special program $\downarrow \in \mathbf{P}$ called *termination*. $\mathbf{F}$ is a set of formulas.

Definition 1 (DL_p Formulas). *A dynamic logical formula ϕ w.r.t. parameters $\mathbf{P}$ and $\mathbf{F}$, called a "parameterized dynamic logic" (DL_p) formula, is defined as follows in BNF form:*

$$\phi =_{df} F \mid \neg\phi \mid \phi \wedge \phi \mid [\alpha]\phi,$$

where $F \in \mathbf{F}$, $\alpha \in \mathbf{P}$. We denote the set of DL_p formulas as $\mathfrak{F}_{dlp}$.

Intuitively, formula $[\alpha]\phi$ means that after all executions of program α, formula ϕ holds. $\langle\cdot\rangle$ is the dual operator of $[\cdot]$. Formula $\langle\alpha\rangle\phi$ is expressed as $\neg[\alpha]\neg\phi$. Other formulas with logical connectives such as $\vee$ and $\rightarrow$ can be expressed by formulas with $\neg$ and $\wedge$ accordingly. Note that in a DL_p formula, there can be multiple and nested modalities, e.g., both $[\alpha]\phi \rightarrow \langle\beta\rangle\psi$ and $[\alpha](\langle\beta\rangle\phi)$ are legal DL_p formulas.

Following the convention of defining a dynamic logic (cf. [21]), we introduce a novel Kripke structure to tackle parameterized program behaviours in $\mathbf{P}$.

Definition 2 (Program-Labelled Kripke Structure). *A "program-labelled" (PL) Kripke structure w.r.t. parameters* $\mathbf{P}$ *and* $\mathbf{F}$ *is a triple*

$$K(\mathbf{P},\mathbf{F}) =_{df} (\mathcal{S}, \longrightarrow, \mathcal{I}),$$

where S *is a set of worlds;* $\longrightarrow \subseteq \mathcal{S} \times (\mathbf{P} \times \mathbf{P}) \times \mathcal{S}$ *is a set of relations labelled by program pairs, in the form of* $w_1 \xrightarrow{\alpha/\alpha'} w_2$ *for some* $w_1, w_2 \in \mathcal{S}$, $\alpha, \alpha' \in \mathbf{P}$*;* $\mathcal{I} : \mathbf{F} \rightarrow \mathcal{P}(\mathcal{S})$ *is an interpretation of formulas in* $\mathbf{F}$ *on the power set of worlds, satisfying that* $w \xrightarrow{\downarrow/\alpha} \cdot$ *for any* $w \in \mathcal{S}$ *and* $\alpha \in \mathbf{P}$ *(capturing the meaning of program* $\downarrow$*).*

PL Kripke structures differ from the Kripke structures of PDL by introducing program-labelled relations $w_1 \xrightarrow{\alpha/\alpha'} w_2$, which describe programs' transitional behaviours. Intuitively, it means that from world w_1, program α is transitioned to program α', ending with world w_2.

Below in this paper, **our discussion is always based on an assumed PL Kripke structure namely** $K(\mathbf{P},\mathbf{F}) = (\mathcal{S}, \longrightarrow, \mathcal{I})$, where transitional behaviours $\longrightarrow$ is usually captured by a set of inference rules on program transitions (see Section 3.2), as the operational semantics of $\mathbf{P}$.

Example 1 (An Instantiation of Programs and Formulas). Consider a while program *WP* in an instantiation namely $\mathbf{P}_W$ of parameter $\mathbf{P}$:

$$WP =_{df} \{\textit{while } (n > 0) \textit{ do } s := s + n \ ; \ n := n - 1 \textit{ end } \}.$$

Given an initial value of variables n and s, program *WP* computes the sum from n to 1 stored in variable s. The underlying theory of programs $\mathbf{P}_W$ is the arithmetic number theory in integer domain $\mathbb{Z}$. Let Var_W be the set of integer variables. $x := e$ is an assignment, meaning assigning the value of expression e to variable x. In the PL Kripke structure $K_W = (\mathcal{S}_W, \longrightarrow, \mathcal{I}_K)$ for $\mathbf{P}_W$, a world $w \in S_W$ is a mapping $w : Var_W \rightarrow \mathbb{Z}$ from variables to integers. Programs' transitional behaviours of $\mathbf{P}_W$ is captured by the relations on K_W. For example, we have a relation $w \xrightarrow{x:=x+1/\downarrow} w[x \mapsto w(x) + 1]$, where $w[x \mapsto v]$ is a mapping that only differs from w on mapping x to value v. The formulas in while programs namely $\mathbf{F}_W$ are the usual arithmetic first-order logical formulas in integer domain $\mathbb{Z}$.

Definition 3 (Execution Path). *An "execution path" on* K *is a finite sequence of relations on* $\longrightarrow$*:* $w_1 \xrightarrow{\alpha_1/\beta_1} \ldots \xrightarrow{\alpha_n/\beta_n} w_{n+1}$ $(n \geq 0)$ *satisfying that* $\beta_n \in \{\downarrow\}$*, and* $\beta_i = \alpha_{i+1} \notin \{\downarrow\}$ *for all* $1 \leq i < n$*.*

In Definition 3, the execution path is sometimes simply written as a sequence of worlds: $w_1...w_{n+1}$. When $n = 0$, the execution path is a single world w_1 (without any relations on $\longrightarrow$).

Definition 4 (Semantics of DL_p Formulas). *Given a DL_p formula ϕ, a world $w \in S$ satisfies ϕ under K, denoted by $K, w \models \phi$, is inductively defined as follows:*

1. $K, w \models F$ *where* $F \in \mathbf{F}$, *if* $w \in \mathcal{I}(F)$;
2. $K, w \models \neg\phi$, *if* $K, w \not\models \phi$;
3. $K, w \models \phi \wedge \psi$, *if* $K, w \models \phi$ *and* $K, w \models \psi$;
4. $K, w \models [\alpha]\phi$, *if for all execution paths of the form:* $w \xrightarrow{\alpha/\cdot} ... \xrightarrow{\cdot/\downarrow} w'$ *for some* $w' \in \mathcal{S}$, $K, w' \models \phi$.

According to the definition of operator $\langle\cdot\rangle$, its semantics is defined such that $K, w \models \langle\alpha\rangle\phi$, if there exists an execution path of the form $w \xrightarrow{\alpha/\cdot} ... \xrightarrow{\cdot/\downarrow} w'$ for some $w' \in \mathcal{S}$ such that $K, w' \models \phi$.

A DL_p formula ϕ is called *valid* w.r.t. K, denoted by $K \models \phi$ (or simply $\models \phi$), if $K, w \models \phi$ for all $w \in \mathcal{S}$.

Example 2 (DL_p Specifications). A property of program *WP* (Example 1) is described as the following formula

$$(n \geq 0 \wedge n = N \wedge s = 0) \rightarrow [WP](s = ((N + 1)N)/2),$$

which means that given an initial condition of n and s, after executing *WP*, s equals to $((N + 1)N)/2$, which is the sum of $1 + 2 + ... + N$, with N a free variable in $\mathbb{Z}$. We will prove an equivalent labelled version of this formula in DL_p in Section 4.

3 A Cyclic Proof System for DL_p

We propose a cyclic proof system for DL_p. We firstly propose a labelled proof system P_{dlp} to support reasoning based on symbolic executions (Section 3.2). Then we construct a cyclic proof structure for system P_{dlp}, which support deriving infinite proof trees under certain conditions (Section 3.3). Section 3.1 introduces the notions of labelled sequent calculus and cyclic proof.

3.1 Prerequisites

Labelled Sequent Calculus. We assume a set $\mathbf{L}$ of labels as a *parameter* of DL_p. A label mapping $\mathfrak{m} : \mathbf{L} \rightarrow \mathcal{S}$ maps each label of $\mathbf{L}$ to a world of set $\mathcal{S}$. Denote the set of all label mappings as $\mathbf{M}$. A *labelled DL_p formula* is of the form $\sigma : \phi$, where $\sigma \in \mathbf{L}$ and $\phi \in \mathfrak{F}_{dlp}$. Denote the set of all labelled DL_p formulas as $\mathfrak{F}_{ldlp}$.

From program-labelled relations defined in Definition 2 we introduce symbolic executions of programs as a type of abstract transitions on labels and program

terminations. A *program transition* is a relation of the form $\sigma \xrightarrow{\alpha/\alpha'} \sigma'$ (also written as $(\alpha, \sigma) \longrightarrow (\alpha', \sigma')$ below) between labels and labelled by a program pair, with $\sigma, \sigma' \in \mathbf{L}$ and $\alpha, \alpha' \in \mathbf{P}$. Call pair (α, σ) a *program state*. We use $\mathfrak{F}_{pt}$ to represent the set of all program transitions. A *program termination* is a relation of the form $\sigma \Downarrow \alpha$ between a label and a program, where $\sigma \in \mathbf{L}$ and $\alpha \in \mathbf{P}$. The set of all program terminations is denoted by $\mathfrak{F}_{ter}$.

A *sequent* is a logical argumentation of the form: $\Gamma \Rightarrow \Delta$, where Γ and Δ are finite multi-sets of formulas, called the *left side* and the *right side* of the sequent respectively. We use dot $\cdot$ to express Γ or Δ when they are empty sets. Intuitively, a sequent $\Gamma \Rightarrow \Delta$ means that if all formulas in Γ hold, then one of formulas in Δ holds. Often use ν to represent a sequent.

A *labelled sequent* is a sequent in which each formula is a formula in $\mathfrak{F}_{ldlp} \cup \mathfrak{F}_{pt} \cup \mathfrak{F}_{ter}$. We often use τ to represent a formula of a labelled sequent.

Definition 5 (Semantics of Formulas in Labelled Sequents). *Given a labelled sequent ν and a label mapping $\mathfrak{m} \in \mathbf{M}$, the satisfaction relation $K, \mathfrak{m} \models \tau$ of a formula τ in ν under K is defined as follows according to different cases of τ:*

1. $K, \mathfrak{m} \models \sigma : \phi$, *if* $K, \mathfrak{m}(\sigma) \models \phi$;
2. $K, \mathfrak{m} \models \sigma \xrightarrow{\alpha/\alpha'} \sigma'$, *if* $\mathfrak{m}(\sigma) \xrightarrow{\alpha/\alpha'} \mathfrak{m}(\sigma')$ *is a relation on* K;
3. $K, \mathfrak{m} \models \sigma \Downarrow \alpha$, *if there exists an execution path* $\mathfrak{m}(\sigma) \xrightarrow{\alpha/\cdot} ... \xrightarrow{\cdot/\downarrow} w$ *on* K *for some world* $w \in \mathcal{S}$.

A formula τ in a labelled sequent is *valid*, denoted by $K \models \tau$ (or simply $\models \phi$), if $K, \mathfrak{m} \models \tau$ for all $\mathfrak{m} \in \mathbf{M}$. According to the meaning of a sequent above, a labelled sequent $\Gamma \Rightarrow \Delta$ is *valid*, if for every $\mathfrak{m} \in \mathbf{M}$, $K, \mathfrak{m} \models \tau$ for all $\tau \in \Gamma$ implies $K, \mathfrak{m} \models \tau'$ for some $\tau' \in \Delta$.

For a multi-set Γ of formulas, we write $K, \mathfrak{m} \models \Gamma$ to mean that $K, \mathfrak{m} \models \tau$ for all $\tau \in \Gamma$.

Example 3 (Instantiation of Labels). In while programs, we consider a type of labels namely $\mathbf{L}_W$ of the form: $\{x_1 \mapsto e_1, ..., x_n \mapsto e_n\}$ $(n \geq 0)$ as program configurations, where each variable $x_i \in Var_W$ stores a unique value of arithmetic expression e_i $(1 \leq i \leq n)$. To make it simple, we restrict that variables $x_1, ...x_n$ must appear in the discussed programs and any free variable in $e_1, ..., e_n$ cannot be any of $x_1, ..., x_n$. For example, in program *WP* (Example 1), $\{n \mapsto N, s \mapsto 0\}$ is a configuration that maps n to value N (as a free variable) and s to 0.

Example 4 (Instantiation of Label Mappings). In while programs, we consider a set $\mathbf{M}_W$ of label mappings where each label mapping is associated to a world, denoted by $\mathfrak{m}_w$ for some $w \in \mathcal{S}_W$. Given a configuration $\sigma =_{df} \{x_1 \mapsto e_1, ..., x_n \mapsto e_n\}$ $(n \geq 1)$, $\mathfrak{m}_w(\sigma)$ is defined as a world such that (1) $\mathfrak{m}_w(\sigma)(x_i) = w(e_i)$ for each x_i $(1 \leq i \leq n)$; (2) $\mathfrak{m}_w(\sigma)(y) = w(y)$ for other variable $y \in Var_W$. Where $w(e_i)$ returns a value by substituting each free variable x of e_i with value $w(x)$. For example, let $w(N) = 5$, then we have $\mathfrak{m}_w(\{n \mapsto N, s \mapsto 0\})(n) = w(N) = 5$, $\mathfrak{m}_w(\{n \mapsto N, s \mapsto 0\})(s) = 0$. In fact, in this case, $\mathfrak{m}_w(\{n \mapsto N, s \mapsto 0\}) = w$.

An *inference rule* is of the form $\frac{\nu_1 \quad ... \quad \nu_n}{\nu}$, where each of ν, ν_i $(1 \leq i \leq n)$ is also called a *node*. Each of $\nu_1, ..., \nu_n$ is called a *premise*, and ν is called the *conclusion*, of the rule. The semantics of the rule is that the validity of sequent $\nu_1, ..., \nu_n$ implies the validity of sequent ν. A formula τ of node ν is called the *target formula* if except τ other formulas are kept unchanged in the derivation from ν to some node ν_i $(1 \leq i \leq n)$. And in this case other formulas except τ in node ν are called the *context* of ν. A formula pair (τ_1, τ_2) with τ_1 in ν and τ_2 in some ν_i is called a *conclusion-premise* (CP) pair of the derivation from ν to ν_i.

Proof & Preproof & Cyclic Proof. A *proof tree* (or *proof*) is a finite tree structure formed by making derivations backwardly from a root node. In a proof tree, a node is called *terminal* if it is the conclusion of an axiom.

In cyclic proof approach (cf. [11]), a *preproof* is an infinite proof tree (i.e. some of its derivations contain infinitely many nodes) in which there exist non-terminal leaf nodes, called *buds*. Each bud is identical to one of its ancestors in the tree. The ancestor identical to a bud ν is called a *companion of* ν. A *derivation path* in a preproof is an infinite sequence of nodes $\nu_1\nu_2...\nu_m...$ $(m \geq 1)$ starting from the root node ν_1, where each node pair (ν_i, ν_{i+1}) $(i \geq 1)$ is a CP pair of a rule. A proof tree is *cyclic*, if it is a preproof in which there exists a "progressive derivation trace", whose definition depends on specific logic theories (see Definition 9 later for DL_p), over every derivation path.

A *proof system* P consists of a finite set of inference rules. We say that a node ν can be derived from P, denoted by $P \vdash \nu$, if a proof tree can be constructed (with ν the root node) by applying the rules in P, which satisfies either (1) all of its leaf nodes terminate or (2) it is a cyclic proof tree.

3.2 A Proof System for DL_p

The labelled proof system P_{dlp} of DL_p is defined as: $P_{dlp} =_{df} P_{ldlp} \cup P_{pt} \cup P_{ter}$, where P_{ldlp} is a set of *core rules* listed in Table 2, P_{pt} and P_{ter} are two finite pre-defined sets of rules according to parameter $\mathbf{P}$.

Sets P_{pt} and P_{ter} capture programs' operational semantics (see Example 5 below). They are for deriving program transitions $\mathfrak{F}_{pt}$ and terminations $\mathfrak{F}_{ter}$ respectively. Each rule in P_{pt} (resp. P_{ter}) has a restricted form: $\frac{\Gamma_1 \Rightarrow \Delta_1 \quad ... \quad \Gamma_n \Rightarrow \Delta_n}{\Gamma \Rightarrow \tau, \Delta}$, where $n \geq 0$, $\tau \in \mathfrak{F}_{pt}$ (resp. $\tau \in \mathfrak{F}_{ter}$), τ is the target formula of the rule.

For sets P_{pt} and P_{ter} to capture the operational semantics of $\mathbf{P}$, assumptions should be made to guarrantee the consistency between system P_{dlp} and PL Kripke structure K. In other words, system P_{dlp} should exactly derive all relations on K.

Definition 6 (Assumptions on P_{dlp}). *System P_{dlp} are assumed to satisfy the following properties:*

1. *Soundness of P_{pt} and P_{ter}. All rules of P_{pt} and P_{ter} are sound.*

$$\frac{}{\Gamma \Rightarrow (x := e, \sigma) \longrightarrow (\downarrow, \sigma_e^x), \Delta}\ (x{:=}e) \qquad \frac{\Gamma \Rightarrow (\alpha_1, \sigma) \longrightarrow (\alpha_1', \sigma'), \Delta}{\Gamma \Rightarrow (\alpha_1; \alpha_2, \sigma) \longrightarrow (\alpha_1'; \alpha_2, \sigma'), \Delta}\ (;)$$

$$\frac{\Gamma \Rightarrow (\alpha_1, \sigma) \longrightarrow (\downarrow, \sigma'), \Delta}{\Gamma \Rightarrow (\alpha_1; \alpha_2, \sigma) \longrightarrow (\alpha_2, \sigma'), \Delta}\ (;\downarrow) \qquad \frac{\Gamma, \sigma : \phi \Rightarrow (\alpha, \sigma) \longrightarrow (\alpha', \sigma'), \Delta \quad \Gamma \Rightarrow \phi : \sigma, \Delta}{\Gamma \Rightarrow (\mathit{while}\ \phi\ \mathit{do}\ \alpha\ \mathit{end}, \sigma) \longrightarrow (\alpha';\ \mathit{while}\ \phi\ \mathit{do}\ \alpha\ \mathit{end}, \sigma'), \Delta}\ (wh1)$$

$$\frac{\Gamma, \sigma : \phi \Rightarrow (\alpha, \sigma) \longrightarrow (\downarrow, \sigma'), \Delta \quad \Gamma \Rightarrow \sigma : \phi, \Delta}{\Gamma \Rightarrow (\mathit{while}\ \phi\ \mathit{do}\ \alpha\ \mathit{end}, \sigma) \longrightarrow (\mathit{while}\ \phi\ \mathit{do}\ \alpha\ \mathit{end}, \sigma'), \Delta}\ (wh1\downarrow) \qquad \frac{\Gamma \Rightarrow \sigma : \neg\phi, \Delta}{\Gamma \Rightarrow (\mathit{while}\ \phi\ \mathit{do}\ \alpha\ \mathit{end}, \sigma) \longrightarrow (\downarrow, \sigma), \Delta}\ (wh2)$$

Table 1. Partial Inference Rules for Program Transitions of While Programs

$$\frac{\{\Gamma \Rightarrow \sigma' : [\alpha']\phi, \Delta\}_{(\alpha', \sigma') \in \Phi}}{\Gamma \Rightarrow \sigma : [\alpha]\phi, \Delta}\ {}^{1}\ ([\alpha]R), \quad \text{where } \Phi =_{df} \{(\alpha', \sigma') \mid P_{dlp} \vdash (\Gamma \Rightarrow \sigma \xrightarrow{\alpha/\alpha'} \sigma', \Delta)\}$$

$$\frac{\Gamma, \sigma' : [\alpha']\phi \Rightarrow \Delta}{\Gamma, \sigma : [\alpha]\phi \Rightarrow \Delta}\ {}^{1}\ ([\alpha]L), \quad \text{if } P_{dlp} \vdash (\Gamma \Rightarrow \sigma \xrightarrow{\alpha/\alpha'} \sigma', \Delta)$$

$$\frac{\sigma : \phi}{\sigma : [\downarrow]\phi}\ ([\downarrow]) \quad \left| \frac{}{\Gamma \Rightarrow \Delta}\ {}^{2}\ (Ter) \right. \quad \left| \frac{\Gamma \Rightarrow \Delta}{Sub(\Gamma) \Rightarrow Sub(\Delta)}\ {}^{3}\ (Sub) \right. \quad \left| \frac{}{\Gamma, \sigma : \phi \Rightarrow \sigma : \phi, \Delta}\ (ax) \right.$$

$$\frac{\Gamma \Rightarrow \sigma : \phi, \Delta \quad \Gamma, \sigma : \phi \Rightarrow \Delta}{\Gamma \Rightarrow \Delta}\ (Cut) \quad \left| \frac{\Gamma \Rightarrow \Delta}{\Gamma \Rightarrow \sigma : \phi, \Delta}\ (WkR) \right. \quad \left| \frac{\Gamma \Rightarrow \Delta}{\Gamma, \sigma : \phi \Rightarrow \Delta}\ (WkL) \right. \quad \left| \frac{\sigma : \phi, \sigma : \phi}{\sigma : \phi}\ (Con) \right.$$

$$\frac{\Gamma, \sigma : \phi \Rightarrow \Delta}{\Gamma \Rightarrow \sigma : \neg\phi, \Delta}\ (\neg R) \quad \left| \frac{\Gamma \Rightarrow \sigma : \phi, \Delta}{\Gamma, \sigma : \neg\phi \Rightarrow \Delta}\ (\neg L) \right. \quad \left| \frac{\Gamma \Rightarrow \sigma : \phi, \Delta \quad \Gamma \Rightarrow \sigma : \psi, \Delta}{\Gamma \Rightarrow \sigma : \phi \wedge \psi, \Delta}\ (\wedge R) \right. \quad \left| \frac{\Gamma, \sigma : \phi, \sigma : \psi \Rightarrow \Delta}{\Gamma, \sigma : \phi \wedge \psi \Rightarrow \Delta}\ (\wedge L) \right.$$

[1] $\alpha \notin \{\downarrow\}$. [2] for each $\sigma : \phi \in \Gamma \cup \Delta$, $\phi \in \mathbf{F}$; Sequent $\Gamma \Rightarrow \Delta$ is valid. [3] *Sub* is given by Definition 7.

Table 2. Rules P_{ldlp} for the Proof System of DL_p

2. *Completeness w.r.t. K. For any* $\sigma \in \mathbf{L}$, Γ *and* $\mathfrak{m} \in \mathbf{M}$ *with* $\mathfrak{m} \models \Gamma$, *if* $\mathfrak{m}(\sigma) \xrightarrow{\alpha/\alpha'} w$ *is a relation on* K *for some* $\alpha, \alpha' \in \mathbf{P}$ *and* $w \in \mathcal{S}$, *then there exists a label* $\sigma' \in \mathbf{L}$ *such that* $\mathfrak{m}(\sigma') = w$ *and* $P_{dlp} \vdash (\Gamma \Rightarrow \sigma \xrightarrow{\alpha/\alpha'} \sigma')$.

Example 5 (Instantiation of $\mathfrak{F}_{pt}$*).* Table 1 displays a set namely $(\mathfrak{F}_{pt})_W$ of partial inference rules for transitions of while programs. Where σ_e^x represents a configuration that store variable x as value e, while storing other variables as the same value as σ.

Through the rules in P_{ldlp}, a labelled DL_p formula can be transformed into proof obligations as non-dynamic formulas, which can then be encoded and verified accordingly through, for example, an SAT/SMT checking procedure. The rules for other operators like $\vee$, $\rightarrow$ can be derived accordingly using the rules in Table 2.

The illustration of each rule in Table 2 is as follows. We use a double-lined inference form: $\dfrac{\phi_1 \quad ... \quad \phi_n}{\phi}$ to represent both rules $\dfrac{\Gamma \Rightarrow \phi_1, \Delta \quad ... \quad \Gamma \Rightarrow \phi_n, \Delta}{\Gamma \Rightarrow \phi, \Delta}$ and $\dfrac{\Gamma, \phi_1 \Rightarrow \Delta \quad ... \quad \Gamma, \phi_n \Rightarrow \Delta}{\Gamma, \phi \Rightarrow \Delta}$, provided any context Γ and Δ.

Rules ($[\alpha]R$) and ($[\alpha]L$) reason about dynamic parts of labelled DL_p formulas. Both rules rely on side deductions: '$P_{dlp} \vdash (\Gamma \Rightarrow \sigma \xrightarrow{\alpha/\alpha'} \sigma', \Delta)$' as sub-proof procedures of program transitions. In rule ($[\alpha]R$), $\{...\}_{(\alpha',\sigma')\in\Phi}$ represents the collection of premises for all program states $(\alpha', \sigma') \in \Phi$. By the finiteness of system P_{dlp}, set Φ must be finite (because only a finite number of forms (α', σ') can be derived). So rule ($[\alpha]R$) only has a finite number of premises. When Φ is empty, the conclusion terminates. Compared to rule ($[\alpha]R$), rule ($[\alpha]L$) has only one premise for some program state (α', σ').

Rule ($[\downarrow]$) deals with the situation when the program is a termination $\downarrow$. Its soundness is straightforward by the semantics of $\downarrow$ in Definition 2.

Rule (*Ter*) indicates that one proof branch terminates when all labelled formulas do not contain any dynamic parts.

Rule (*Sub*) describes a specialization process for labelled DL_p formulas. For a set A of labelled formulas, $Sub(A) =_{df} \{Sub(\sigma) : \phi \mid (\sigma : \phi) \in A\}$, with *Sub* an abstract notion of substitution defined as follows in Definition 7. Intuitively, if sequent $\Gamma \Rightarrow \Delta$ is valid, then its one of special cases $Sub(\Gamma) \Rightarrow Sub(\Delta)$ is also valid. Rule (*Sub*) plays an important role in constructing a bud in a cyclic proof structure (Section 3.3). See Section 4 for more details.

Definition 7 (Substitution of Labels). *A 'substitution' $\eta : \mathbf{L} \to \mathbf{L}$ is a function on $\mathbf{L}$ satisfying that for any label mapping $\mathfrak{m} \in \mathbf{M}$, there exists a label mapping $\mathfrak{m}'(\mathfrak{m}, \eta)$ (determined only by $\mathfrak{m}$ and η) such that $\mathfrak{m}'(\sigma) = \mathfrak{m}(\eta(\sigma))$ for all labels $\sigma \in \mathbf{L}$.*

Definition 7 will be used in the proof of soundness of rules P_{ldlp} and the cyclic proof system of DL_p.

Rules from (ax) to ($\wedge L$) are the "labelled verions" of the corresponding rules inherited from traditional first-order logic. Their meanings are classical and we omit their discussions here.

Theorem 1. *Each rule from P_{ldlp} in Table 2 is sound.*

Following the above explanations, Theorem 1 can be proved according to the semantics of labelled DL_p formulas under the assumption of Definition 6. See Appendix A for more details.

3.3 Construction of A Cyclic Proof Structure of DL_p

In an ordinary proof system we usually expect a finite proof tree. However, in system P_{dlp}, a branch of a proof tree does not always terminate. Because the process of symbolically executing a program via rule ($[\alpha]R$) or/and rule ($[\alpha]L$) might not stop. This is well known when a program has an explicit/implicit loop structure that may run infinitely. For example, a while program $W =_{df}$ *while true do* $x := x + 1$ *end* will proceed infinitely as the following program transitions: $(W, \{x \mapsto 0\}) \longrightarrow (W, \{x \mapsto 1\}) \longrightarrow ....$

In this paper, we build a cyclic labelled proof system for DL_p, in order to recognize and admit potential infinite derivations as above when deriving labelled

DL_p formulas $\mathfrak{F}_{ldlp}$ using rules in P_{ldlp}. Based on the notion of preproof (Section 3.1), we build a cyclic proof structure for system P_{dlp}, where the key part is to introduce the notion of progressive derivation traces in DL_p (Definition 9). Section 5 will further show that a cyclic proof of DL_p ensures a valid conclusion.

Next we first introduce the notion of progressive derivation traces for DL_p, then we define the cyclic proof structure for DL_p, as a special case of the notion already given in Section 3.1.

Definition 8 (Derivation Traces). *A "derivation trace" over a derivation path $\mu_1\mu_2...\mu_k\nu_1\nu_2...\nu_m...$ ($k \geq 0, m \geq 1$) is an infinite sequence $\tau_1\tau_2...\tau_m...$ of formulas with each formula τ_i ($1 \leq i \leq m$) in node ν_i. Each CP pair (τ_i, τ_{i+1}) ($i \geq 1$) of derivation (ν_i, ν_{i+1}) satisfies special conditions as follows according to (ν_i, ν_{i+1}) being the different instances of rules from P_{ldlp}:*

1. *If (ν_i, ν_{i+1}) is an instance of rule $([\alpha]R)$, $([\alpha]L)$, $([\downarrow])$, $(\neg R)$, $(\neg L)$, $(\wedge R)$ or $(\wedge L)$, then either τ_i is the target formula and τ_{i+1} is its replacement of the rule, or $\tau_i = \tau_{i+1}$;*
2. *If (ν_i, ν_{i+1}) is an instance of rule (Sub), then $\tau_i = Sub(\sigma) : \phi$ and $\tau_{i+1} = \sigma : \phi$ for some $\sigma \in \mathbf{L}$ and $\phi \in \mathfrak{F}_{dlp}$;*
3. *If (ν_i, ν_{i+1}) is an instance of other rules, then $\tau_i = \tau_{i+1}$.*

Definition 9 (Progressive Derivation Traces). *In a preproof of system P_{dlp}, given a derivation trace $\tau_1\tau_2...\tau_m...$ over a derivation path $...\nu_1\nu_2...\nu_m...$ ($m \geq 1$) starting from τ_1 in node ν_1, a CP pair (τ_i, τ_{i+1}) ($1 \leq i \leq m$) of derivation (ν_i, ν_{i+1}) is called a "progressive step", if (τ_i, τ_{i+1}) is the following CP pair of an instance of rule $([\alpha]R)$:*

$$\frac{... \quad \nu_{i+1} :: (\Gamma \Rightarrow \tau_{i+1} :: (\sigma' : [\alpha']\phi), \Delta) \quad ...}{\nu_i :: (\Gamma \Rightarrow \tau_i :: (\sigma : [\alpha]\phi), \Delta),} \ ([\alpha]R);$$

or the following CP pair of an instance of rule $([\alpha]L)$:

$$\frac{\nu_{i+1} :: (\Gamma, \tau_{i+1} :: (\sigma' : [\alpha']\phi) \Rightarrow \Delta)}{\nu_i :: (\Gamma, \tau_i :: (\sigma : [\alpha]\phi) \Rightarrow \Delta)} \ ([\alpha]L),$$

provided with an additional side deduction $P_{dlp} \vdash (\Gamma \Rightarrow \sigma' \Downarrow \alpha', \Delta)$.

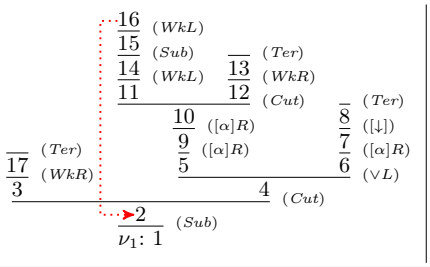
If a derivation trace has an infinite number of progressive steps, we say that the trace is 'progressive'.

The additional side condition of the instance of rule $([\alpha]L)$ is the key to prove the corresponding case in Lemma 1 (see Appendix A).

Theorem 2. *Every cyclic proof of system P_{dlp} has a sound conclusion.*

In Section 5, we will analyze and prove Theorem 2 under a restriction on $\mathbf{P}$.

Since DL_p is not a specific logic, it is impossible to discuss about its decidability, completeness or whether it is cut-free without any restrictions on parameters $\mathbf{P}, \mathbf{F}$ and $\mathbf{L}$. One of our future work will focus on analyzing under what restrictions, these properties can be obtained in a general sense.



Definitions of other symbols:
$WP =_{df} \{while\ (n > 0)\ do\ s := s + n\ ;\ n := n - 1\ end\ \}$
$\alpha_1 =_{df}\ s := s + n\ ;\ n := n - 1$
$\phi_1 =_{df}\ (s = ((N+1)N)/2)$
$\sigma_1 =_{df} \{n \mapsto N, s \mapsto 0\}$
$\sigma_2 =_{df} \{n \mapsto N - m, s \mapsto (2N - m + 1)m/2\}$
$\sigma_3 =_{df} \{n \mapsto N - m, s \mapsto (2N - (m+1) + 1)(m+1)/2\}$
$\sigma_4 =_{df} \{n \mapsto N - (m+1), s \mapsto (2N - (m+1) + 1)(m+1)/2\}$

1:	$\sigma_1 : n \geq 0$	$\Rightarrow$	$\sigma_1 : [while\ (n > 0)\ do\ \alpha_1\ end\ ]\phi_1$
2:	$\sigma_2 : n \geq 0$	$\Rightarrow$	$\sigma_2 : [while\ (n > 0)\ do\ \alpha_1\ end\ ]\phi_1$
3:	$\sigma_2 : n \geq 0$	$\Rightarrow$	$\sigma_2 : [while\ (n > 0)\ do\ \alpha_1\ end\ ]\phi_1, \sigma_2 : (n > 0 \vee n \leq 0)$
18:	$\sigma_2 : n \geq 0$	$\Rightarrow$	$\sigma_2 : (n > 0 \vee n \leq 0)$
4:	$\sigma_2 : n \geq 0, \sigma_2 : (n > 0 \vee n \leq 0)$	$\Rightarrow$	$\sigma_2 : [while\ (n > 0)\ do\ \alpha_1\ end\ ]\phi_1$
5:	$\sigma_2 : n \geq 0, \sigma_2 : n > 0$	$\Rightarrow$	$\sigma_2 : [while\ (n > 0)\ do\ \alpha_1\ end\ ]\phi_1$
9:	$\sigma_2 : n \geq 0, \sigma_2 : n > 0$	$\Rightarrow$	$\sigma_3 : [n := n - 1;\ while\ (n > 0)\ do\ \alpha_1\ end\ ]\phi_1$
10:	$\sigma_2 : n \geq 0, \sigma_2 : n > 0$	$\Rightarrow$	$\sigma_4 : [while\ (n > 0)\ do\ \alpha_1\ end\ ]\phi_1$
11:	$\sigma_2 : n \geq 0, \sigma_2 : n > 0, \sigma_4 : n \geq -1, \sigma_4 : n \geq 0$	$\Rightarrow$	$\sigma_4 : [while\ (n > 0)\ do\ \alpha_1\ end\ ]\phi_1$
14:	$\sigma_4 : n \geq -1, \sigma_4 : n \geq 0$	$\Rightarrow$	$\sigma_4 : [while\ (n > 0)\ do\ \alpha_1\ end\ ]\phi_1$
15:	$\sigma_2 : n \geq -1, \sigma_2 : n \geq 0$	$\Rightarrow$	$\sigma_2 : [while\ (n > 0)\ do\ \alpha_1\ end\ ]\phi_1$
16:	$\sigma_2 : n \geq 0$	$\Rightarrow$	$\sigma_2 : [while\ (n > 0)\ do\ \alpha_1\ end\ ]\phi_1$
12:	$\sigma_2 : n \geq 0, \sigma_2 : n > 0$	$\Rightarrow$	$\sigma_4 : [while\ (n > 0)\ do\ \alpha_1\ end\ ]\phi_1, \sigma_4 : n \geq -1, \sigma_4 : n \geq 0$
13:	$\sigma_2 : n \geq 0, \sigma_2 : n > 0$	$\Rightarrow$	$\sigma_4 : n \geq -1, \sigma_4 : n \geq 0$
6:	$\sigma_2 : n \geq 0, \sigma_2 : n \leq 0$	$\Rightarrow$	$\sigma_2 : [while\ (n > 0)\ do\ \alpha_1\ end\ ]\phi_1$
7:	$\sigma_2 : n \geq 0, \sigma_2 : n \leq 0$	$\Rightarrow$	$\sigma_2 : [\downarrow]\phi_1$
8:	$\sigma_2 : n \geq 0, \sigma_2 : n \leq 0$	$\Rightarrow$	$\sigma_2 : (s = ((N+1)N)/2)$

Table 3. Derivations of Property ν_1

4 Case Study

4.1 A Cyclic Deduction for While Programs

We give an example to show how a DL_p formula can be derived according to rules in Table 2. We prove the property in Example 2, which can be captured by the following equivalent labelled sequent

$$\nu_1 =_{df}\ \sigma_1 : n \geq 0 \Rightarrow \sigma_1 : [WP](s = ((N+1)N)/2),$$

where $\sigma_1 =_{df} \{n \mapsto N, s \mapsto 0\}$, describing the initial configuration of WP.

Table 3 shows its derivations. We omit all side deductions as sub-proof procedures in instances of rule $([\alpha]R)$ derived using the inference rules in Table 1. Non-primitive rule $(\vee L)$ can be derived by the rules for $\neg$ and $\wedge$ accordingly.

The derivation from sequent 1 to 2 is according to rule (Sub), where the substitution Sub of labels defined in Definition 7 is instantiated as function $(\cdot)[e/x]$. Informally, for any label σ, $\sigma[e/x]$ returns the label by substituting each free variable x of σ with term e. We observe that $\sigma_1 = \sigma_2[0/m]$, so sequent 1 is a special case of sequent 2 by substitution $(\cdot)[0/m]$. Intuitively, label σ_2 captures

$$\frac{}{\Gamma \Rightarrow \mathcal{U} \xrightarrow{x:=e/\downarrow} \{x := e\}\mathcal{U}, \Delta}\ (x{:=}e) \qquad \frac{\Gamma \Rightarrow \mathcal{U} \xrightarrow{\alpha/\alpha'} \mathcal{U}', \Delta}{\Gamma \Rightarrow \mathcal{U} \xrightarrow{\alpha\,;\beta/\alpha';\beta} \mathcal{U}', \Delta}\ 1\ (;) \qquad \frac{\Gamma \Rightarrow \mathcal{U} \xrightarrow{\alpha/\downarrow} \mathcal{U}', \Delta}{\Gamma \Rightarrow \mathcal{U} \xrightarrow{\alpha\,;\beta/\beta} \mathcal{U}', \Delta}\ (;\downarrow) \qquad \frac{\Gamma \Rightarrow \mathcal{U} \xrightarrow{\alpha/\alpha'} \mathcal{U}', \Delta}{\Gamma \Rightarrow \mathcal{U} \xrightarrow{\alpha \cup \beta/\alpha'} \mathcal{U}', \Delta}\ (\cup 1)$$

$$\frac{\Gamma \Rightarrow \mathcal{U} \xrightarrow{\beta/\beta'} \mathcal{U}', \Delta}{\Gamma \Rightarrow \mathcal{U} \xrightarrow{\alpha \cup \beta/\beta'} \mathcal{U}', \Delta}\ (\cup 2) \qquad \frac{\Gamma \Rightarrow \mathcal{U} \xrightarrow{\alpha/\alpha'} \mathcal{U}', \Delta}{\Gamma \Rightarrow \mathcal{U} \xrightarrow{\alpha^*/\alpha';\alpha^*} \mathcal{U}', \Delta}\ 2\ (*1) \qquad \frac{\Gamma \Rightarrow \mathcal{U} \xrightarrow{\alpha/\downarrow} \mathcal{U}' \Delta}{\Gamma \Rightarrow \mathcal{U} \xrightarrow{\alpha^*/\alpha^*} \mathcal{U}', \Delta}\ 2\ (*2) \qquad \frac{}{\Gamma \Rightarrow \mathcal{U} \xrightarrow{\alpha^*/\downarrow} \mathcal{U}, \Delta}\ (*\downarrow)$$

Table 4. Inference Rules $(P_{pt})_{FO}$ for Program Transitions of Regular Programs

the program configuration after the mth loop ($m \geq 0$) of program *WP*. This step is crucial as starting from sequent 2, we can find a bud node — 16 — that is identical to node 2.

The derivation from sequent 2 to {3, 4} provides a lemma: $\sigma_2 : (n > 0 \vee n \leq 0)$, which is obvious valid. Sequent 16 indicates the end of the $(m + 1)$th loop of program *WP*. From node 10 to 16, we transform the formulas on the left side into an obvious logical equivalent form in order to apply rule (*Sub*) from sequent 14 to 15. Sequent 14 a special case of sequent 15 since $\sigma_4 = \sigma_2[m + 1/m]$.

The whole proof tree is cyclic because the only derivation path: $2, 4, 5, 9, 10, 11, 14, 15, 16, 2, ...$ has a progressive derivation trace whose elements are underlined in Table 3.

Compared to the deduction processes in traditional dynamic logics and Hoare logics, a notable feature of the above deduction process is that the search for a loop invariant is reflected in a suitable configuration (i.e. σ_2). One advantage of this approach is that sometimes configuration structures might be more intuitive and easier to construct.

As a demonstration of its powerfulness, in Appendix B of [1], we briefly introduce another instantiation of DL_p for Esterel [8] and show a cyclic derivation of an Esterel program in which we use rule $([\alpha]L)$ to perform symbolic executions instead of rule $([\alpha]R)$. That example can better highlight the advantages brought by the cyclic proof framework of DL_p since the loop structure there is implicit. And where we also show that how using symbolic-based reasoning in DL_p for synchronous languages can benefit from avoiding unnecessary language transformations.

4.2 Instantiation of FODL in DL_p

We show that FODL can be instantiated in the theory of DL_p. As FODL is the basic theory underlying many dynamic-logic variations (such as [5,34,48,33]), it demonstrates how DL_p can be potentially useful in specifying and reasoning about different languages.

Below we only give an informal instantiation, as these structures are prevalent and can be found in previous work such as [21,4].

Let $\mathbf{P}_{FO}$ be the set of regular programs of FODL and $\mathbf{F}_{FO}$ be the set of non-dynamic FODL formulas. $K_{FO} = (\mathcal{S}_{FO}, \longrightarrow, \mathcal{I}_{FO})$ is the PL Kripke structure of FODL, where each world $w \in \mathcal{S}_{FO}$ is a mapping $w : Var_{FO} \to \mathbb{Z}$ from the set Var_{FO} of variables to integer domain.

An 'update' is of the form: $\{x := e\}$ (cf. [4]) where $x \in Var_{FO}$ and e is an expression. For a formula ϕ, an update $\{x := e\}$ on ϕ, denoted by $\{x := e\}(\phi)$, returns a formula by substituting each free variable x of ϕ with e. A label $\mathcal{U} \in \mathbf{L}_{FO}$ of FODL is a sequence of 'updates', which applies to a formula ϕ by applying each update of σ on ϕ from right to left, denoted by $\mathcal{U}(\phi)$. For example, an application $\{x := y\}\{x := x + 1\}(x = 5)$ results in $y + 1 = 5$. When a label $\mathcal{U}$ contains no free variables, it is a world in $\mathcal{S}$.

For a world $w \in K_{FO}$, let $w(\mathcal{U})$ return an update by substituting each free variable x of $\mathcal{U}$ with value $w(x)$. So $w(\mathcal{U})$ (without free variables) is a world itself. Therefore, each world w itself is a label mapping, and we have $\mathbf{M}_{FO} = \mathcal{S}$ in K_{FO}.

Transitional behaviours of K_{FO} are captured by the inference rules namely $(P_{pt})_{FO}$ in Table 4 following Definition 6. One can also define a set $(P_{ter})_{FO}$ of inference rules for the terminations of regular programs, which we omit in this paper.

5 Soundness of Cyclic Proof System P_{dlp}

In this section, we prove Theorem 2 under a restriction on $\mathbf{P}$ given as in Definition 10.

An execution path (Definition 3) $w_1...w_n$ $(n \geq 1)$ is called *minimum*, if there are no two relations $w_i \xrightarrow{\alpha_i/\cdot} \cdot$ and $w_j \xrightarrow{\alpha_j/\cdot} \cdot$ for some $1 \leq i < j < n$ such that $w_i = w_j$ and $\alpha_i = \alpha_j$. Intuitively, in a minimum execution path, there are no two relations starting from the same world and program.

Definition 10 (Termination Finiteness). *In PL Kripke structure K, for a world $w \in \mathcal{S}$ and a program $\alpha \in \mathbf{P}$, there is only a finite number of minimum execution paths starting from a relation $w \xrightarrow{\alpha/\cdot} \cdot$ for some $w \in \mathcal{S}$ and $\alpha \in \mathbf{P}$.*

Programs satisfying termination finiteness are in fact a rich set, including, for example, all deterministic programs, such as while programs discussed in this paper, programming languages like Esterel, C, Java, etc. Many non-deterministic programs also fall into this category, especially those that have the same expressiveness as regular expressions with tests of FODL in domain $\mathbb{Z}$. More on this restriction will be discussed in our future work.

Main Idea. We follow the main idea behind [10] to prove Theorem 2 by contradiction. The key point is that, if the conclusion of a cyclic proof is invalid, then there must exist an *invalid derivation path* in which each node is invalid, and one of its progressive traces would lead to an infinite descent sequence of some well-founded set (introduced below), which violates the definition of the well-foundedness itself.

Below we firstly introduce the well-founded relation $\prec_m$ we will rely on, then we focus on the main skeleton of proving Theorem 2. Other proof details are given in the Appendix A.

Well-foundedness & Relation $\preceq_m$. Given a set S and a partial-order relation $\preceq$ on S, $\preceq$ is called a *well-founded relation* over S, if for any element a in S, there is no infinite descent sequence: $a \succ a_1 \succ a_2 \succ ...$ in S. Set S is called a *well-founded* set w.r.t. $\preceq$.

Definition 11 (Relation $\preceq_m$). *Given two finite sets C_1 and C_2 of finite execution paths, $C_1 \preceq_m C_2$ is defined if either (1) $C_1 = C_2$; or (2) set C_1 can be obtained from C_2 by replacing (or removing) one or more elements of C_2 each with a finite number of elements, such that for each replaced element tr, its replacements $tr_1, ..., tr_n$ ($n \geq 1$) in C_1 are proper suffixes of tr.*

$\preceq_m$ is a partial-order relation. Its proof is given in Appendix A.

Example 6. Let $C_1 = \{tr_1, tr_2, tr_3\}$, where $tr_1 =_{df} ww_1w_2w_3w_4, tr_2 =_{df} ww_1w_5w_6w_7$ and $tr_3 =_{df} ww_8$; $C_2 = \{tr'_1, tr'_2\}$, where $tr'_1 =_{df} w_1w_2w_3w_4, tr'_2 =_{df} w_1w_5w_6w_7$. We see that tr'_1 is a proper suffix of tr_1 and tr'_2 is a proper suffix of tr_2. C_2 can be obtained from C_1 by replacing tr_1 and tr_2 with tr'_1 and tr'_2 respectively, and removing tr_3. Hence $C_2 \preceq_m C_1$. Since $C_1 \neq C_2$, $C_2 \prec_m C_1$.

Proposition 1. *Relation $\preceq_m$ is a well-founded relation.*

We omit the proof of Proposition 1. Relation $\preceq_m$ is just a special case of the "multi-set ordering" introduced in [13], where it has been proved to be well-founded.

Proof Skeleton of Theorem 2. Below we give the main skeleton of the proof by skipping the details of the proof of Lemma 1, which can be found in Appendix A.

Following the main idea above, we first introduce the concept of "execution paths of a dynamic DL_p formula". It is as the elements of a well-founded relation $\preceq_m$. Next, we propose Lemma 1, which plays a key role in the proof of Theorem 2 that follows.

Given two execution paths $w_1...w_n$ and $w'_1w'_2...w'_m$ ($n, m \geq 1$), *path concatenation* $\cdot$ is a partial function defined such that $(w_1...w_n) \cdot (w'_1w'_2...w'_m) =_{df} w_1...w_nw'_2...w'_m$, if $w_n = w'_1$.

Definition 12 (Execution Paths of Dynamic Formulas). *Given a world $w \in \mathcal{S}$ and a dynamic formula ϕ, the execution paths $EX(w, \phi)$ of ϕ w.r.t. w is inductively defined according to the structure of ϕ as follows:*

1. $EX(w, [\alpha]F) =_{df} mex(w, \alpha)$, *where* $F \in \mathbf{F}$;
2. $EX(w, [\alpha]\phi_1) =_{df} mex(w, \alpha) \cup \{tr_1 \cdot tr_2 \mid tr_1 \in mex(w, \alpha), tr_2 \in EX((tr_1)_e, \phi_1)\}$;
3. $EX(w, \neg\phi_1) =_{df} EX(w, \phi_1)$;
4. $EX(w, \phi_1 \wedge \phi_2) =_{df} EX(w, \phi_1) \cup EX(w, \phi_2)$.

Where $mex(w, \alpha) =_{df} \{w...w' \mid w \xrightarrow{\alpha/\cdot} ... \xrightarrow{\cdot/\downarrow} w'$ *is a min. exec. path for some* $w' \in \mathcal{S}\}$ *is the set of all minimum paths of α starting from world w.*

We call $\mathfrak{m} \in \mathbf{M}$ a *counter-example mapping* of a node ν, if it makes ν invalid.

Lemma 1. *In a cyclic proof (where there is at least one derivation path), let* $(\sigma : \phi, \sigma' : \phi')$ *be a step of a derivation trace over a derivation* (ν, ν') *of an invalid derivation path, where* $\phi, \phi' \in \mathfrak{F}_{dlp}$. *For any set* $EX(\mathfrak{m}(\sigma), \phi)$ *of* $\sigma : \phi$ *w.r.t. a counter-example mapping* $\mathfrak{m}$ *of* ν, *there exists a counter-example mapping* $\mathfrak{m}'$ *of* ν' *and a set* $EX(\mathfrak{m}'(\sigma'), \phi')$ *of* $\sigma' : \phi'$ *such that* $EX(\mathfrak{m}'(\sigma'), \phi') \preceq_m EX(\mathfrak{m}(\sigma), \phi)$. *Moreover, if* $(\sigma : \phi, \sigma' : \phi')$ *is a progressive step, then* $EX(\mathfrak{m}'(\sigma'), \phi') \prec_m EX(\mathfrak{m}(\sigma), \phi)$.

Intuitively, Lemma 1 helps us discover suitable execution-path sets imposed by a well-founded relation $\preceq_m$ between them in an invalid derivation path.

Based on Proposition 1 and Lemma 1, we give the proof of Theorem 2 as follows.

Proof (Proof of Theorem 2). By contradiction. Let the progressive trace be $\tau_1\tau_2...\tau_k...$ over a derivation path $...\nu_1\nu_2...\nu_k...$ $(k \geq 1)$ starting from τ_1 in ν_1, where $\tau_i =_{df} \sigma_i : \phi_i$ $(i \geq 1)$, each ν_i $(i \geq 1)$ is invalid.

Since ν_1 is invalid, let $\mathfrak{m}_1$ be one of its counter-example mappings. By Lemma 1, from $EX(\mathfrak{m}_1(\sigma_1), \phi_1)$, there exists an infinite sequence of sets $EX_1, ..., EX_k, ...$ $(k \geq 1)$, where each $EX_i =_{df} EX(\mathfrak{m}_i(\sigma_i), \phi_i)$ $(i \geq 1)$ with $\mathfrak{m}_i$ a counter-example mapping of node ν_i, and which satisfies that $EX_1 \succeq_m ... \succeq_m EX_k \succeq_m ....$ Moreover, since trace $\tau_1\tau_2...\tau_k...$ is progressive (Definition 9), there must be an infinite number of $j \geq 1$ such that $EX_j \succ_m EX_{j+1}$. This thus forms an infinite descent sequence w.r.t. $\prec_m$, violating the well-foundedness of relation $\preceq_m$ (Proposition 1).

6 Related Work

The idea of reasoning about programs based on their operational semantics is not new. Previous work such as [40,41,43,12] in last decade has addressed this issue using theories based on rewriting logic [29]. Matching logic [40] is based on patterns and pattern matching. Its basic form, a reachability rule $\varphi \Rightarrow \varphi'$ (where $\Rightarrow$ means differently from in sequents here), captures whether pattern φ' is reachable from pattern φ in a given pattern reachability system. Based on matching logic, one-path and all-paths reachability logics [41,43] were developed by enhancing the expressive power of the reachability rule. A more powerful matching μ-logic [12] was proposed by adding a least fixpoint μ-binder to matching logic.

In these theories, 'patterns' is a more general structure. So to encode the dynamic forms $[\alpha]\phi$ in DL_p requires additional work and program transformations. On the other hand, dynamic logics like DL_p are natural to express and reason about complex before-after and temporal program properties with their modalities $[\cdot]$ and $\langle\cdot\rangle$. In terms of expressiveness, matching logic and one-path reachability logic cannot capture the semantics of modality $[\cdot]$ when the programs are non-deterministic (which means more than one execution paths). Because in their theories the reachability rule $\varphi \Rightarrow \varphi'$ only considers the existence of one execution path. We conjecture that matching μ-logic can encode DL_p, as it has been claimed that it can encode traditional dynamic logics (cf. [12]).

[30] proposed a general program verification framework based on coinduction. Using the terminology in this paper, a program specification $\sigma : [\alpha]\phi$ can be expressed as a pair $((\alpha, \sigma), P(\phi))$ in [30], with $P(\phi)$ a set of program states capturing the semantics of formula ϕ. A method was designed to derive a program specification in a coinductive way according to the operational semantics of (α, σ). Following [30], [28] also proposed a general framework for program reasoning, but via big-step operational semantics. Unlike the frameworks in [30] and [28] which are directly built up on mathematical set theory, DL_p is in logical forms, and is based on a cyclic deduction approach rather than coinduction. In terms of expressiveness, the meaning of modality $\langle\cdot\rangle$ in DL_p cannot be expressed in the framework of [30]. More detailed comparisons should be given between coinduction and cyclic deduction in our future work.

The structure 'updates' adopted in work [34,4,5] are "delay substitutions" of variables and terms. They in fact can be defined as a special case of the more general labels in DL_p by choosing suitable label mappings accordingly.

The proof system of DL_p relies on the cyclic proof theory which firstly arose in [44] and was later developed in different logics such as [11] and [10]. [26] proposed a complete cyclic proof system for μ-calculus, which subsumes PDL [17] in its expressiveness. In [15], the authors proposed a complete labelled cyclic proof system for PDL. Both μ-calculus and PDL only study valid formulas in all domains. They can never tell a formulas in particular domains, as the example shown in DL_p. The labelled form of DL_p formula $\sigma : [\alpha]\phi$ is inspired from [15], where a label is just a variable of worlds in a traditional Kripke structure. While the labels in DL_p allow arbitrary terms from actual program configurations.

There has been some other work for generalizing the theories of dynamic logics, such as [31,22]. However, generally speaking, they neither consider structures as general as allowed by programs and configurations in DL_p, nor adopt a similar approach for reasoning about programs. [31] proposed a complete generic dynamic logic for monads of programming languages. In the dynamic logic proposed in [22], more general programs can be reasoned about than regular expressions of PDL, based on so-called "interaction-based" behaviours. But there program transitions are captured by abstract actions in a form e.g. $\alpha \xrightarrow{a} \beta$, where no explicit structures of program configurations are allowed. And yet no proof systems have been built for that logic.

7 Conclusion & Future Work

In this paper, we propose a novel dynamic logic DL_p that supports reasoning about general forms of programs and formulas based on programs' operational semantics. We mainly build the theory of DL_p and propose a sound cyclic proof system of DL_p that is proved to be useful and applicable. As the main theoretical result we prove the soundness of DL_p. One future work will be a full mechanization of DL_p using Coq [9]. To see the full potential of DL_p, we are also interested in instantiating more types of programs or system models in DL_p.

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A Other Propositions and Proofs

Lemma 2. *Given labelled formulas $\tau_1, ..., \tau_n, \tau$, for any label mapping $\mathfrak{m} \in \mathbf{M}$, if $K, \mathfrak{m} \models \tau_1$ and ... and $K, \mathfrak{m} \models \tau_n$ implies $K, \mathfrak{m} \models \tau$, then rule*

$$\frac{\Gamma \Rightarrow \tau_1, \Delta \quad ... \quad \Gamma \Rightarrow \tau_n, \Delta}{\Gamma \Rightarrow \tau, \Delta}$$

is sound for any contexts Γ, Δ.

Proof (Proof of Lemma 2). Assume $\Gamma \Rightarrow \tau_1, \Delta$,...,$\Gamma \Rightarrow \tau_n, \Delta$ are valid, for any $\mathfrak{m} \in \mathbf{M}$, let $K, \mathfrak{m} \models \Gamma$ but let $K, \mathfrak{m} \not\models \tau'$ for all $\tau' \in \Delta$, we need to prove $K, \mathfrak{m} \models \tau$. From the assumption we have $K, \mathfrak{m} \models \tau_1$, ..., $K, \mathfrak{m} \models \tau_n$. Then $K, \mathfrak{m} \models \tau$ is an immediate result.

Lemma 3. *Given labelled formulas $\tau_1, ..., \tau_n, \tau$, for any label mapping $\mathfrak{m} \in \mathbf{M}$, if $K, \mathfrak{m} \models \tau$ implies either $K, \mathfrak{m} \models \tau_1$ or ... or $K, \mathfrak{m} \models \tau_n$, then rule*

$$\frac{\Gamma, \tau_1 \Rightarrow \Delta \quad ... \quad \Gamma, \tau_n \Rightarrow \Delta}{\Gamma, \tau \Rightarrow \Delta}$$

is sound for any contexts Γ, Δ.

Proof (Proof of Lemma 3). Assume $\Gamma, \tau_1 \Rightarrow \Delta$,...,$\Gamma, \tau_n \Rightarrow \Delta$ are valid. For any $\mathfrak{m} \in \mathbf{M}$, if $K, \mathfrak{m} \models \Gamma$ and $K, \mathfrak{m} \models \tau$, by the assumption, $K, \mathfrak{m} \models \tau_i$ for some $1 \leq i \leq n$. By the validity of $\Gamma, \tau_i \Rightarrow \Delta$, $K, \mathfrak{m} \models \Delta$. By the arbitrariness of $\mathfrak{m}$ we know that $\Gamma, \tau \Rightarrow \Delta$ is valid.

Content of Theorem 1: Each rule from P_{ldlp} in Table 2 is sound.

Below we only prove the soundness of rules ($[\alpha]R$), ($[\alpha]L$) and (*Sub*). Other rules can be proved similarly based on the semantics of DL_p formulas (Definition 4).

Proof (Proof of Theorem 1).

For rule ($[\alpha]R$), by Lemma 2, it is sufficient to prove that for any $\mathfrak{m} \in \mathbf{M}$ with $\mathfrak{m} \models \Gamma$, if $K, \mathfrak{m} \models \sigma' : [\alpha']\phi$ for all $(\alpha', \sigma') \in \Phi$, then $K, \mathfrak{m} \models \sigma : [\alpha]\phi$. For any relation $\mathfrak{m}(\sigma) \xrightarrow{\alpha/\alpha_0} w$ with some $\alpha_0 \in \mathbf{P}$ and $s_0 \in S$, by the completeness w.r.t. $\mathfrak{F}_{pt}$ of system P_{dlp} (Definition 6), there is a label $\sigma_0 \in \mathbf{L}$ such that $\mathfrak{m}(\sigma_0) = s_0$ and $P_{dlp} \vdash (\Gamma \Rightarrow \sigma \xrightarrow{\alpha/\alpha_0} \sigma_0)$. So $P_{dlp} \vdash (\Gamma \Rightarrow \sigma \xrightarrow{\alpha/\alpha_0} \sigma_0, \Delta)$ (by rule (*WkR*)) and $(\alpha_0, \sigma_0) \in \Phi$. Hence $K, \mathfrak{m} \models \sigma_0 : [\alpha_0]\phi$. By the arbitrariness of α_0 and σ_0, we have $K, \mathfrak{m} \models \sigma : [\alpha]\phi$ according to the semantics of formula $[\alpha]\phi$ (Definition 4).

For rule ($[\alpha]L$), by Lemma 3, it is sufficient to prove that for any $\mathfrak{m} \in \mathbf{M}$ with $\mathfrak{m} \models \Gamma$, if $K, \mathfrak{m} \models \sigma : [\alpha]\phi$, then $K, \mathfrak{m} \models \sigma' : [\alpha']\phi$. For any execution path $\mathfrak{m}(\sigma')...w$ of α', by the soundness of the side deduction $P_{dlp} \vdash (\Gamma \Rightarrow \sigma \xrightarrow{\alpha/\alpha} \sigma', \Delta)$, $\mathfrak{m}(\sigma) \xrightarrow{\alpha/\alpha'} \mathfrak{m}(\sigma')$ is a relation on K, so $\mathfrak{m}(\sigma)\mathfrak{m}(\sigma')...w$ is an execution path of α. By the semantics of formula $[\alpha]\phi$ (Definition 4), we obtain the result.

For rule (*Sub*), we need to prove that the validity of $\Gamma \Rightarrow \Delta$ implies the validity of $Sub(\Gamma) \Rightarrow Sub(\Delta)$. For any $\mathfrak{m} \in \mathbf{M}$ satisfying $K, \mathfrak{m} \models Sub(\Gamma)$, by Definition 7, there exists a $\mathfrak{m}'$ such that for any formula $\sigma : \phi \in \Gamma \cup \Delta$, $\mathfrak{m}'(\sigma) = \mathfrak{m}(Sub(\sigma))$. So $K, \mathfrak{m}' \models \Gamma$. Since $\Gamma \Rightarrow \Delta$ is valid, $K, \mathfrak{m}' \models \sigma_0 : \phi_0$ for some $\sigma_0 : \phi_0 \in \Delta$. By that $\mathfrak{m}'(\sigma_0) = \mathfrak{m}(Sub(\sigma_0))$, $K, \mathfrak{m} \models Sub(\sigma_0) : \phi_0$ with $Sub(\sigma_0) : \phi_0 \in Sub(\Delta)$. By the arbitrariness of $\mathfrak{m}$, $Sub(\Gamma) \Rightarrow Sub(\Delta)$ is valid.

Proposition 2. *$\preceq_m$ is a partial-order relation.*

Before proving Proposition 2, we firstly review the notion of *partial-order relation*.

A relation $\preceq$ on a set S is *partially ordered*, if it satisfies the following properties: (1) Reflexivity. $t \preceq t$ for each $t \in S$. (2) Anti-symmetry. For any $t_1, t_2 \in S$, if $t_1 \preceq t_2$ and $t_2 \preceq t_1$, then t_1 and t_2 are the same element in S, we denote by $t_1 = t_2$. (3) Transitivity. For any $t_1, t_2, t_3 \in S$, if $t_1 \preceq t_2$ and $t_2 \preceq t_3$, then $t_1 \preceq t_3$. $t_1 \prec t_2$ is defined as $t_1 \preceq t_2$ and $t_1 \neq t_2$.

Below we use $\preceq_s$ to represent the suffix relation between execution paths, which is obviously a partial-order relation.

Proof (Proof of Proposition 2). The reflexivity is trivial. The transitivity can be proved by the definition of 'replacements' as described in Definition 11 and the transitivity of relation $\prec_s$. Below we only prove the anti-symmetry.

For any finite sets D_1, D_2 of finite paths, if $D_1 \preceq_m D_2$ but $D_1 \neq D_2$, let $f_{D_1,D_2} : D_1 \to D_2$ be the function defined such that for any $tr \in D_1$, either (1) $f_{D_1,D_2}(tr) \approx_s tr$; or (2) tr is one of the replacements of a replaced element $f_{D_1,D_2}(tr)$ in D_2 with $tr \prec_s f_{D_1,D_2}(tr)$.

For the anti-symmetry, suppose $C_1 \preceq_m C_2$ and $C_2 \preceq_m C_1$ but $C_1 \neq C_2$. Let $tr \in C_1$ but $tr \notin C_2$. Then from $C_1 \preceq_m C_2$, we have $tr \prec_s f_{C_1,C_2}(tr)$. Since $f_{C_1,C_2}(tr)$ is the replaced element, so $f_{C_1,C_2}(tr) \notin C_1$. By $C_2 \preceq_m C_1$, we have $f_{C_1,C_2}(tr) \prec_s f_{C_2,C_1}(f_{C_1,C_2}(tr))$. Continuing this process, we can construct an infinite descent sequence $tr \prec_s f_{C_1,C_2}(tr) \prec_s f_{C_2,C_1}(f_{C_1,C_2}(tr)) \prec_s ...$ w.r.t. relation $\preceq_s$, which violates its well-foundedness. So the only possibility is $C_1 = C_2$.

Content of Lemma 1: In a cyclic proof (where there is at least one derivation path), let $(\sigma : \phi, \sigma' : \phi')$ be a step of a derivation trace over a derivation (ν, ν') of an invalid derivation path, where $\phi, \phi' \in \mathfrak{F}_{dlp}$. For any set $EX(\mathfrak{m}(\sigma), \phi)$ of $\sigma : \phi$ w.r.t. a counter-example mapping $\mathfrak{m}$ of ν, there exists a counter-example mapping $\mathfrak{m}'$ of ν' and a set $EX(\mathfrak{m}'(\sigma'), \phi')$ of $\sigma' : \phi'$ such that $EX(\mathfrak{m}'(\sigma'), \phi') \preceq_m EX(\mathfrak{m}(\sigma), \phi)$. Moreover, if $(\sigma : \phi, \sigma' : \phi')$ is a progressive step, then $EX(\mathfrak{m}'(\sigma'), \phi') \prec_m EX(\mathfrak{m}(\sigma), \phi)$.

Proof (Proof of Lemma 1). Consider the rule application from node ν, we only consider the cases when it is an instance of rule $([\alpha]R)$, rule $([\alpha]L)$, rule (Sub), and rule $(\wedge R)$.

Case for rule $([\alpha]R)$**:** If from node ν rule $([\alpha]R)$ is applied with $\tau =_{df} \sigma : [\alpha]\phi$ the target formula, let $\tau' = (\sigma' : [\alpha']\phi)$ for some α' and σ', so $\nu' = (\Gamma \Rightarrow \tau', \Delta)$. Since $\mathfrak{m}$ is a counter-example of ν, $mex(\mathfrak{m}(\sigma), \alpha) \neq \emptyset$, thus $EX(\mathfrak{m}(\sigma), [\alpha]\phi) \neq \emptyset$. By the soundness of rule $([\alpha]R)$ and the assumption that $\vdash_M \Gamma \Rightarrow \sigma \xrightarrow{\alpha/\alpha'} \sigma', \Delta$, for each path $tr = \mathfrak{m}(\sigma')s_1...s_n \in EX(\mathfrak{m}(\sigma'), [\alpha']\phi)$ $(n \geq 0)$, path $\mathfrak{m}(\sigma)tr \in EX(\mathfrak{m}(\sigma), [\alpha]\phi)$ has tr as its proper suffix. By Definition 10 ("termination finiteness"), $EX(\mathfrak{m}(\sigma), [\alpha]\phi)$ and $EX(\mathfrak{m}(\sigma'), [\alpha']\phi)$ are also finite. Therefore $EX(\mathfrak{m}(\sigma'), [\alpha']\phi) \prec_m EX(\mathfrak{m}(\sigma), [\alpha]\phi)$.

Case for rule $([\alpha]L)$**:** If from node ν rule $([\alpha]L)$ is applied with $\tau =_{df} \sigma : [\alpha]\phi$ the target formula, let $\tau' = (\sigma' : [\alpha']\phi)$ for some α' and σ', so $\nu' =$

$(\Gamma \Rightarrow \tau', \Delta)$. By the soundness of rule $([\alpha]L)$ and the assumption that $P_{dlp} \vdash \Gamma \Rightarrow \sigma \xrightarrow{\alpha/\alpha'} \sigma', \Delta$, for each path $tr = \mathfrak{m}(\sigma')s_1...s_n \in EX(\mathfrak{m}(\sigma'), [\alpha']\phi)$ $(n \geq 0)$, path $\mathfrak{m}(\sigma)tr \in EX(\mathfrak{m}(\sigma), [\alpha]\phi)$ has tr as its proper suffix. By Definition 10 ("termination finiteness"), $EX(\mathfrak{m}(\sigma), [\alpha]\phi)$ and $EX(\mathfrak{m}(\sigma'), [\alpha']\phi)$ are also finite.

Now consider two cases: (1) If $mex(\mathfrak{m}(\sigma), \alpha) = \emptyset$, then by the above $mex(\mathfrak{m}(\sigma'), \alpha')$ is also empty. So by the definition of EX, $EX(\rho, \tau', \nu') = EX(\rho, \tau, \nu)$; (2) If $mex(\mathfrak{m}(\sigma), \alpha) \neq \emptyset$, especially, when (τ, τ') is a progressive step with the support of the derivation $P_{dlp} \vdash (\Gamma \Rightarrow \alpha' \Downarrow \sigma', \Delta))$, then $EX(\mathfrak{m}(\sigma), [\alpha]) \neq \emptyset$, moreover, we have $EX(\mathfrak{m}(\sigma'), [\alpha']\phi) \prec_m EX(\mathfrak{m}(\sigma), [\alpha]\phi)$.

Case for rule $(\boldsymbol{Sub})$**:** If from node ν a substitution rule (Sub) is applied, let $\tau = Sub(\sigma) : \phi$ be the target formula of ν, then $\tau' = \sigma : \phi$. By Definition 7, for the label mapping $\mathfrak{m}$ and the substitution Sub, there exists a $\mathfrak{m}'$ such that $\mathfrak{m}'(\sigma) = \mathfrak{m}(Sub(\sigma))$. So $EX(\mathfrak{m}, Sub(\sigma), \phi) = EX(\mathfrak{m}', \sigma, \phi)$.

Case for rule $(\wedge R)$**:** If from node ν a substitution rule $(\wedge R)$ is applied, let $\tau = \sigma : \phi \wedge \psi$ be the target formula of ν, and $\tau' = \sigma : \phi$. By the definition of EX, we have $EX(\mathfrak{m}(\sigma), \phi) \subseteq EX(\mathfrak{m}(\sigma), \sigma : \phi \wedge \psi)$, so $EX(\mathfrak{m}(\sigma), \phi) \preceq_m EX(\mathfrak{m}(\sigma), \sigma : \phi \wedge \psi)$.

From the case of rule $([\alpha]L)$ above, we see that derivation $P_{dlp} \vdash (\Gamma \Rightarrow \alpha' \Downarrow \sigma', \Delta))$ imposes $EX(m(\sigma), [\alpha]\phi) \neq \emptyset$, which is the key to prove the strict relation '$EX(\mathfrak{m}(\sigma'), [\alpha']\phi) \prec_m EX(\mathfrak{m}(\sigma), [\alpha]\phi)$'.

B A Synchronous Program and Its Cyclic Deduction

In this section, we introduce another example of instantiations of DL_p and display its cyclic deductions in system P_{dlp}. We show how system P_{dlp} manages to verify a program whose "loop structures" are implicit. That is where DL_p is really useful as the loop information can be tracked in the cyclic proof structures of DL_p. This example also depicts that how synchronous languages can benefit from symbolic-based reasoning in order to avoid additional program transformations.

B.1 An Esterel Program in DL_p

We consider a synchronous program E of an instantiation $\mathbf{P}_E$ of programs written in Esterel language [8] as follows:

$$E =_{df} \{trap\ A \parallel B\ end\},$$

where

$A =_{df} \{loop\ (emit\ S(0)\ ;\ x := x - S\ ;\ if\ x = 0\ then\ exit\ end\ ;\ pause)\ end\}$

$B =_{df} \{loop\ (emit\ S(1)\ ;\ pause)\ end\}.$

Different from while programs, in program E, $x, S \in Var_E$ are two variables of different types. x is a "local variable", with $\mathbb{Z}$ as its domain. Signal S is a variable called 'signal' whose domain is $\mathbb{Z} \cup \{\bot\}$, with $\bot$ to express the absence

World w	$w(x)$	$w(S)$	nth Instance
w_1	3	1	1
w_2	2	1	2
w_3	1	1	3
w_4	0	1	4

Table 5. Transitional Behaviours of Program E Starting From w_1

of a signal. We assume a PL Kripke structure $K_E = (\mathcal{S}_E, \longrightarrow, \mathcal{I}_E)$ for $\mathbf{P}_E$, where each world $w \in \mathcal{S}_E$ is a mapping $w : Var_E \to (\mathbb{Z} \cup \{\bot\})$ that maps each local variable to an integer and maps each signal to a value of $\mathbb{Z} \cup \{\bot\}$.

A program configuration $\sigma \in \mathbf{L}_E$ in $\mathbf{P}_E$, as a label, is of the form $\{x_1 \mapsto e_1 \,|\, ... \,|\, x_n \mapsto e_n\}$ $(n \geq 1)$, where the only difference from that of a while program is that it is a stack (with $x_n \mapsto e_n$ the top element), allowing several local variables with the same name. For example, configuration $\{x \mapsto 5 \,|\, y \mapsto 1 \,|\, y \mapsto 2\}$ has two different local variables y, storing the values 1 and 2 respectively.

Given a world $w \in \mathcal{S}_E$, a label mapping $\mathfrak{m}_w \in \mathbf{M}_E$ (w.r.t. w) is defined such that for any configuration $\sigma \in \mathbf{L}_E$ of the form: $\{x_1 \mapsto e_1 \,|\, ... \,|\, x_n \mapsto e_n\}$, $\mathfrak{m}_w(\sigma)$ is a world satisfying that (1) $\mathfrak{m}_w(\sigma)(x_i) = w(e_j)$ with $n \geq j \geq i \geq 1$ and j the largest index such that $x_i \mapsto e_j$ in σ (i.e. the right-most value of variable x_i); (2) $\mathfrak{m}_w(\sigma)(y) = w(y)$ for other variable $y \in Var_E$, where $w(e)$ for an expression e is defined similarly as in while programs (see Example 4). For example, $\mathfrak{m}_w(\{x \mapsto 5 \,|\, y \mapsto 1 \,|\, y \mapsto 2\})(y) = w(2) = 2$ for any $\mathfrak{m}_w$.

Different from while programs, the behaviour of a synchronous program is characterized by a sequence of *instances*. At each instance, several (atomic) executions of a program may occur. The value of each variable is unique in an instance. When several programs run in parallel, their executions at one instance are thought to occur simultaneously. In this manner, the behaviour of a parallel synchronous program is deterministic.

In this example, the key word *pause* marks the end of an instance. At the beginning of each instance, the values of all signals are set to $\bot$, representing state 'absence'. The *loop* statements in programs A and B are executed repeatedly for an infinite number of times until some *exit* statement is encountered. Signal emission *emit* $S(e)$ means assigning the value of an expression e to a signal S and broadcasts the value. At each instance, program A firstly emits signal S with value 0 and subtracts x with the current value of S; then checks if $x = 0$. While program B emits signal S with value 1. The value of signal S in one instance should be the sum of all of its emitted values by different current programs. So the value of S should be $1 + 0 = 1$. The whole program E continues executing until $x = 0$ is satisfied, when *exit* terminates the whole program by jumping out of the *trap* statement.

For instance, starting from an initial world w_1 with $w_1(x) = 3$ and $w_1(S) = 1$, we have an execution path $w_1w_2w_3w_4$ of program E explained in Table 5, where we omit the intermediate forms of programs during the execution.

In imperative synchronous programming languages, the behaviour of a parallel program is usually not true interleaving. There exist data dependencies between concurrent programs, which makes its reasoning non-compositional. For instance, in the program E above, the assignment $x := x - S$ can only be executed after all values of S in both programs A and B are collected. In other words, $x := x - S$ can only be executed after *emit* $S(0)$ and *emit* $S(1)$. For synchronous programs like this, additional program transformations are mandatory. For example, in [18], a Quartz program often needs to be firstly transformed into a canonical form called "synchronous tuple assignment", in order to perform a compositional reasoning in Hoare logic. As will be seen next in Section B.2, our approach in the verification framework of DL_p provides a more direct reasoning based on symbolic executions compared to [18].

B.2 Cyclic Deduction of Program *E*

We prove the property

$$\nu_2 =_{df} \sigma_1 : x > 0 \Rightarrow \sigma_1 : \langle E \rangle true,$$

which says that under configuration $\sigma_1 = \{x \mapsto v, S \mapsto \bot\}$, with v a free variable representing an initial value of x, if $v > 0$, then E can finally terminate (satisfying *true*).

The derivations of ν_2 is depicted in Table 6. The symbolic executions of program E rely on rule

$$\frac{\Gamma \Rightarrow \sigma' : \langle \alpha' \rangle \phi, \Delta}{\Gamma \Rightarrow \sigma : \langle \alpha \rangle \phi, \Delta} \ (\langle \alpha \rangle), \text{ if } P_{dlp} \vdash (\Gamma \Rightarrow \sigma \xrightarrow{\alpha/\alpha'} \sigma', \Delta),$$

which can be derived by rule $([\alpha]L)$ and rules $(\neg R)$ and $(\neg L)$ from Table 2. We omit all side deductions as sub-proof procedures in instances of rule $(\langle \alpha \rangle)$, which depends on the operational semantics of Esterel programs (cf. [8]).

From node 2 to 3 is a progressive step, where we omit the derivations of the termination $\sigma_3 \Downarrow trap\ ((x := x - S\ ;A')\ ;\ A) \parallel (pause\ ;\ B)\ end$, which depends on the operational semantics of Esterel. Intuitively, this program does terminate as the value of variable x decreases by 1 (by executing $x := x - S$) in each loop so that statement *exit* will finally be reached. From node 13 to 14 and node 15 to 16, rule

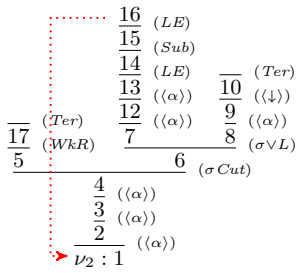
$$\frac{\Gamma, \phi' \Rightarrow \Delta}{\Gamma, \phi \Rightarrow \Delta} \ (LE), \text{ if } \phi \rightarrow \phi' \in \mathbf{F} \text{ is valid}$$

is applied, which can be derived by the following derivations:

$$\frac{\dfrac{\Gamma, \phi' \Rightarrow \Delta}{\Gamma, \phi, \phi' \Rightarrow \Delta}\ (WkL) \qquad \dfrac{}{\Gamma, \phi \Rightarrow \phi', \Delta}\ (Ter)}{\Gamma, \phi \Rightarrow \Delta}\ (Cut)$$

From node 14 to 15, rule (Sub)

$$\frac{\Gamma \Rightarrow \Delta}{\Gamma[e/x] \Rightarrow \Delta[e/x]}\ (Sub)$$



Definitions of other symbols:
$A =_{df}$ *loop* (*emit* $S(0)$; $x := x - S$; *if* $x = 0$ *then exit end* ; *pause*) *end*
$B =_{df}$ *loop* (*emit* $S(1)$; *pause*) *end*
$A' =_{df}$ (*if* $x = 0$ *then exit end* ; *pause*)
$\sigma_1 =_{df} \{x \mapsto v \,|\, S \mapsto \bot\}$
$\sigma_2 =_{df} \{x \mapsto v \,|\, S \mapsto 0\}$
$\sigma_3 =_{df} \{x \mapsto v \,|\, S \mapsto 1\}$
$\sigma_4 =_{df} \{x \mapsto v - 1 \,|\, S \mapsto 1\}$
$\sigma_5 =_{df} \{x \mapsto v - 1 \,|\, S \mapsto \bot\}$

1: $\sigma_1 : x > 0$	$\Rightarrow \underline{\sigma_1 : \langle trap\ A \parallel B\ end \rangle true}$
2: $\sigma_1 : x > 0$	$\Rightarrow \underline{\sigma_2 : \langle trap\ ((x := x - S\ ; A')\ ;\ A) \parallel B)\ end \rangle true}$
3: $\sigma_1 : x > 0$	$\Rightarrow \underline{\sigma_3 : \langle trap\ ((x := x - S\ ; A')\ ;\ A) \parallel (pause\ ;\ B)\ end \rangle true}$
4: $\sigma_1 : x > 0$	$\Rightarrow \underline{\sigma_4 : \langle trap\ (A'\ ;\ A) \parallel (pause\ ;\ B)\ end \rangle true}$
5: $\sigma_1 : x > 0$	$\Rightarrow \sigma_4 : \langle trap\ (A'\ ;\ A) \parallel (pause\ ;\ B)\ end \rangle true, \sigma_1 : (x - 1 \neq 0 \vee x - 1 = 0)$
17: $\sigma_1 : x > 0$	$\Rightarrow \sigma_1 : (x - 1 \neq 0 \vee x - 1 = 0)$
6: $\sigma_1 : x > 0, \sigma_1 : (x - 1 \neq 0 \vee x - 1 = 0)$	$\Rightarrow \underline{\sigma_4 : \langle trap\ (A'\ ;\ A) \parallel (pause\ ;\ B)\ end \rangle true}$
7: $\sigma_1 : x > 0, \sigma_1 : x - 1 \neq 0$	$\Rightarrow \underline{\sigma_4 : \langle trap\ (A'\ ;\ A) \parallel (pause\ ;\ B)\ end \rangle true}$
12: $\sigma_1 : x > 0, \sigma_1 : x - 1 \neq 0$	$\Rightarrow \underline{\sigma_4 : \langle trap\ (pause\ ;\ A) \parallel (pause\ ;\ B)\ end \rangle true}$
13: $\sigma_1 : x > 0, \sigma_1 : x - 1 \neq 0$	$\Rightarrow \underline{\sigma_5 : \langle trap\ A \parallel B\ end \rangle true}$
14: $\sigma_5 : x + 1 > 0, \sigma_5 : x \neq 0$	$\Rightarrow \underline{\sigma_5 : \langle trap\ A \parallel B\ end \rangle true}$
15: $\sigma_1 : x + 1 > 0, \sigma_1 : x \neq 0$	$\Rightarrow \underline{\sigma_1 : \langle trap\ A \parallel B\ end \rangle true}$
16: $\sigma_1 : x > 0$	$\Rightarrow \underline{\sigma_1 : \langle trap\ A \parallel B\ end \rangle true}$
8: $\sigma_1 : x > 0, \sigma_1 : x - 1 = 0$	$\Rightarrow \sigma_4 : \langle trap\ (A'\ ;\ A) \parallel (pause\ ;\ B)\ end \rangle true$
9: $\sigma_1 : x > 0, \sigma_1 : x - 1 = 0$	$\Rightarrow \sigma_5 : \langle \downarrow \rangle true$
10: $\sigma_1 : x > 0, \sigma_1 : x - 1 = 0$	$\Rightarrow \sigma_5 : true$

Table 6. Derivations of Property ν_2

is applied, with $(\cdot)[e/x]$ an instantiation of the substitution *Sub* of labels (Definition 7), defined just as that in the example of Section 4.1. Observe that sequent 14 can be written as:

$$\sigma_1[v-1/v] : x + 1 > 0, \sigma_1[v-1/v] : x \neq 0 \Rightarrow \sigma_1[v-1/v] : \langle trap\ A \parallel B\ end \rangle true.$$

Sequent 16 is a bud with sequent 1 as its companion. The whole preproof is cyclic as the only derivation path: $1, 2, 3, 4, 6, 7, 12, 13, 14, 15, 16, 1, ...$ has a progressive trace whose elements are underlined in Table 6.